



Switch on the d-c supply by closing the DPST switch. Using the three point starter, start the motor. Run the motor at synchronous speed by varying the motor field rheostat. Close the TPST switch. By adjusting the autotransformer, apply 20% to 30% of the rated voltage to the armature of the synchronous machine. Make sure that the direction of rotation of the prime mover and the direction of rotation of the magnetic field produced in the armature are in the same direction by closing the SPST switch. If the voltmeter reading across the alternator field winding is very small, both the directions are correct. If the voltmeter reading is high, interchange the two lines of 3 phase supply after switching off the 3 phase supply. SPST switch is kept open.

The speed is slightly reduced/increased from synchronous speed, so that slip is increased and the voltmeter and ammeter readings are oscillating. The maximum and minimum readings of voltmeter and ammeter are noted. The above said procedure can be repeated with two more different autotransformer settings.

(During slip test, it would be observed that swing of the ammeter pointer is very wide. whereas the voltmeter has only small swing because of the low impedance voltage drop in the leads and 3-phase autotransformer).